Loan Default Analysis

Identifying risk factors for a loan company

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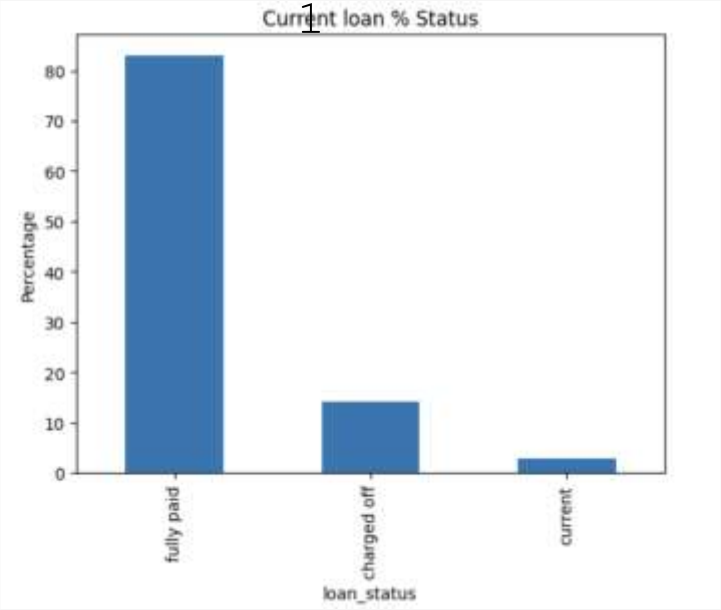
Project Background

- This project aims to understand the strategy of finance company which specialises in lending loans.
- Using EDA, we first understand the dataset, assess if there are an biases or imbalances.
- We then analyse the variables in the dataset, using univariate, bivariate and multivariate analysis.

Business Problem

- The company wants to uncover the characteristics of its borrowers
- Applicants who are more likely to default
- Hence plan out the decision modelling and limit risk.

Fig. 1



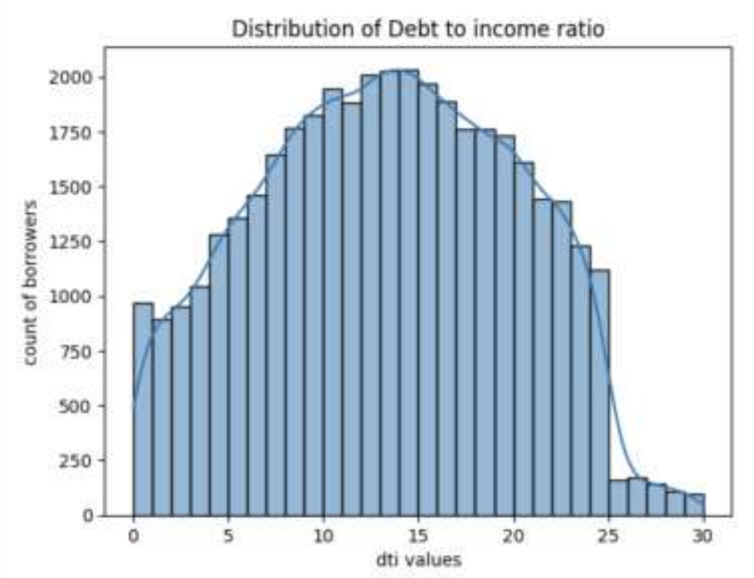


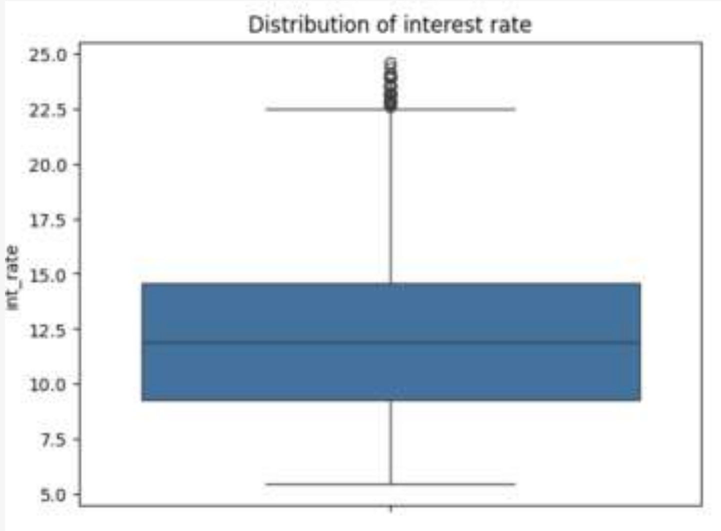
Fig. 2

Univariate Analysis

A. Most important factor to identify risks is by analysing loan status. Graph in Fig1 shows 14% are of loans are charged off i.e. defaulted

B. Debt to income ration also gives a good insight in the kind of borrowers the loan company. Plot in Fig 2. indicates, DTI goes down at the end meaning very less borrowers are risky, which is common as lenders don't approve loans for people with high DTI ratio

Fig.



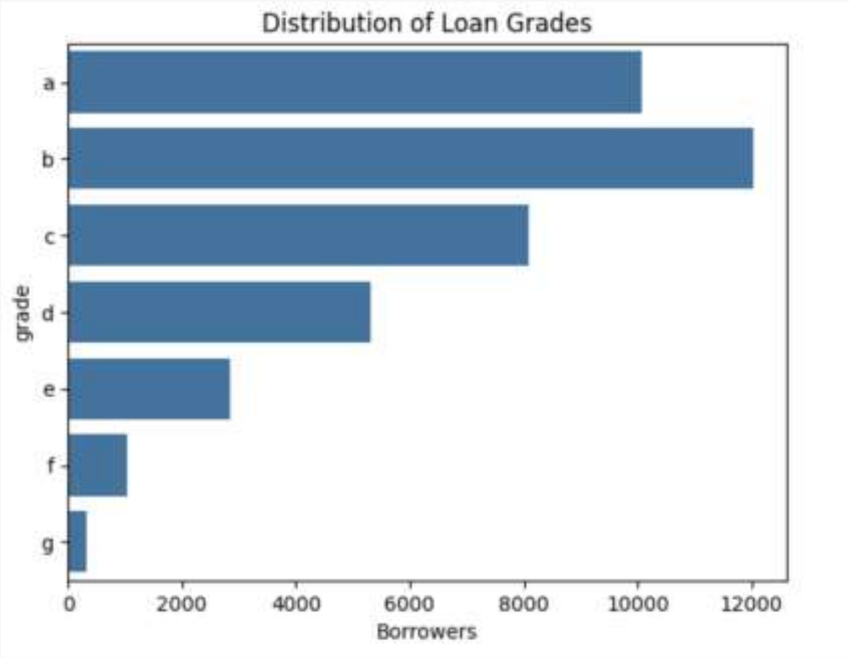
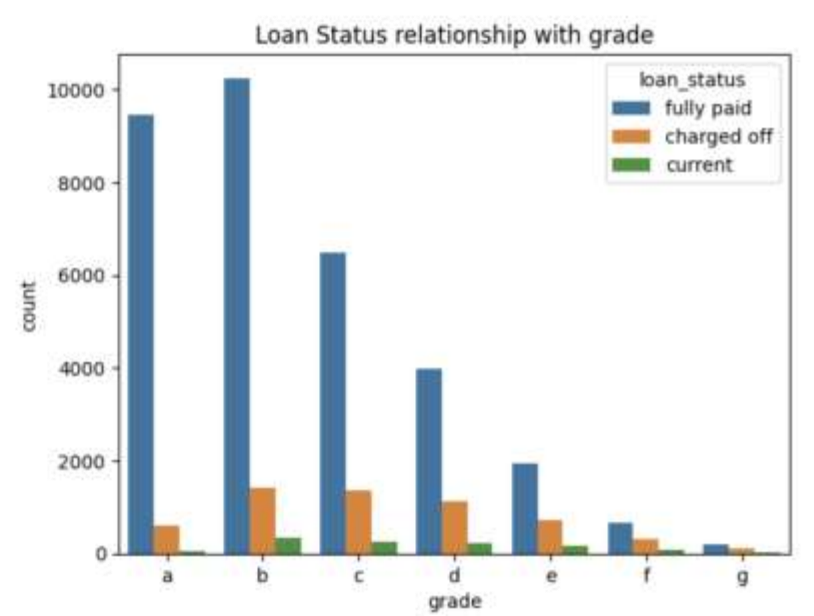


Fig. 4

C. Next variable to understand is interest rate as that tells a story of risk zones within a loan company. Fig3, shows us that 50% of loans fall between 10-15 interest rate, hence its the most typical interest rate given by our Loan company

D. Fig4 gives us the picture of how the grades are structured within the company. Using categorical variable 'grade' here, we can drew an analysis that company gives most of the loans to higher grades applicants i.e. a-c which is a low-risk strategy

Fig.



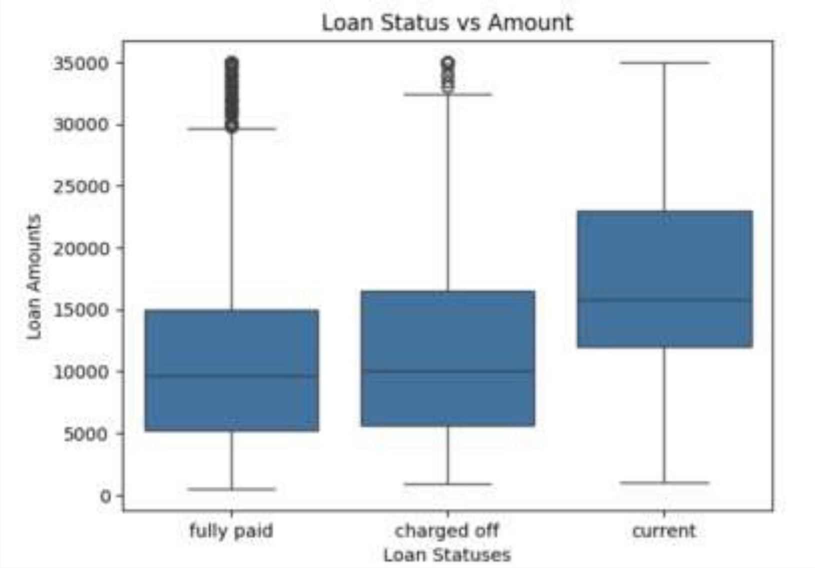


Fig.
2

Bivariate Analysis

- Each grade in the data set reflects a risk category assigned by the Lending Company. Here,
  Grade A-C = Low risk borrowers
  Grade D-G = High risk borrowers. As we can see in the Fig 1, low risk borrowers are getting smaller amounts, and higher grades getting larger loans. It means company is controlling exposure to risky applicants

- To understand the defaulters, we have used boxplot and fig 2 shows the applicants who defaulted (charged off), had higher interest rates

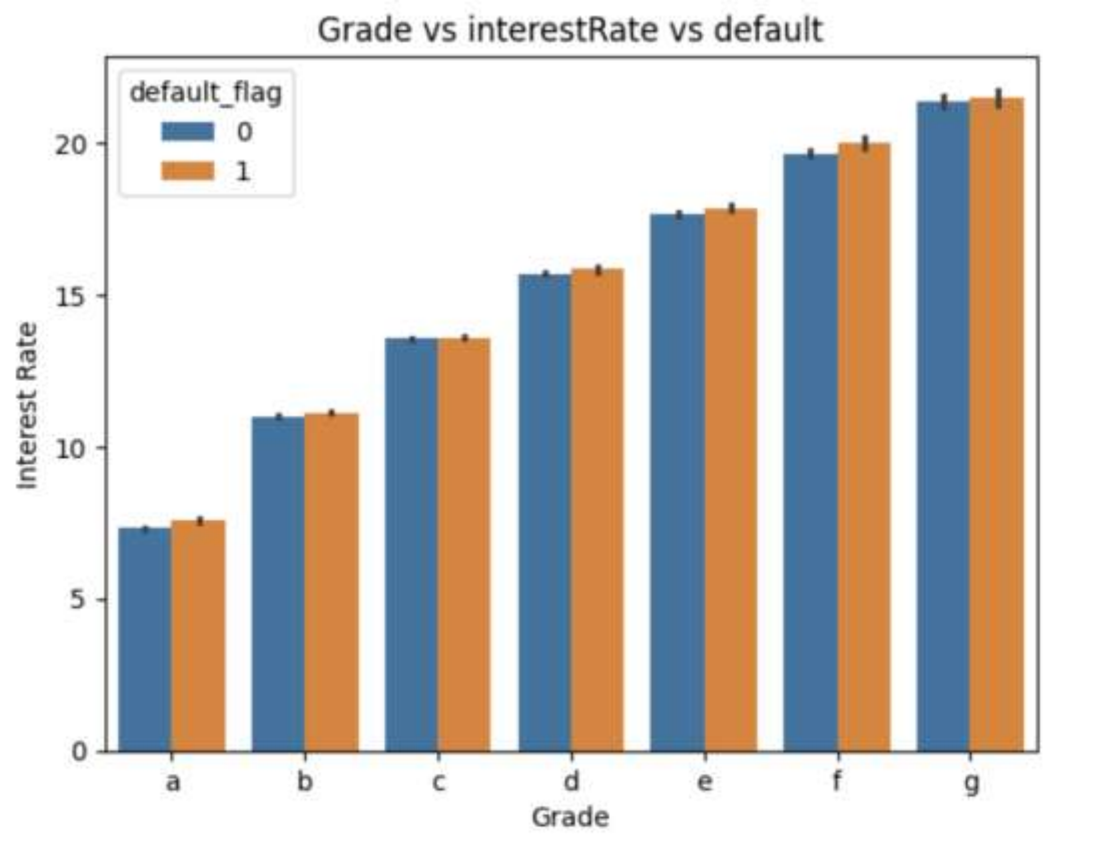


Fig. 1

Multivariate Analysis

Here, using tri-variable analysis on Grade, interest rate, default flag (1- Defaulted, 0- Fully Paid) to get the most insightful relationship in our dataset

So using the boxplot (fig1) we are examining how interest rates vary with grades and how that affect default rates

As you move from grade a-> g, the interest rate steadily increases. This outlook tells the company's risk based pricing

High risk borrowers = higher rates to compensate for that risk

Second observation, even after adjusting for risk via higher interest rates, defaulters are still rising as we go towards lower grades. This means higher interest rate strain borrowers repayment potential.

Conclusion based on this- higher risk borrowers need tighter approval and not just higher interest rates

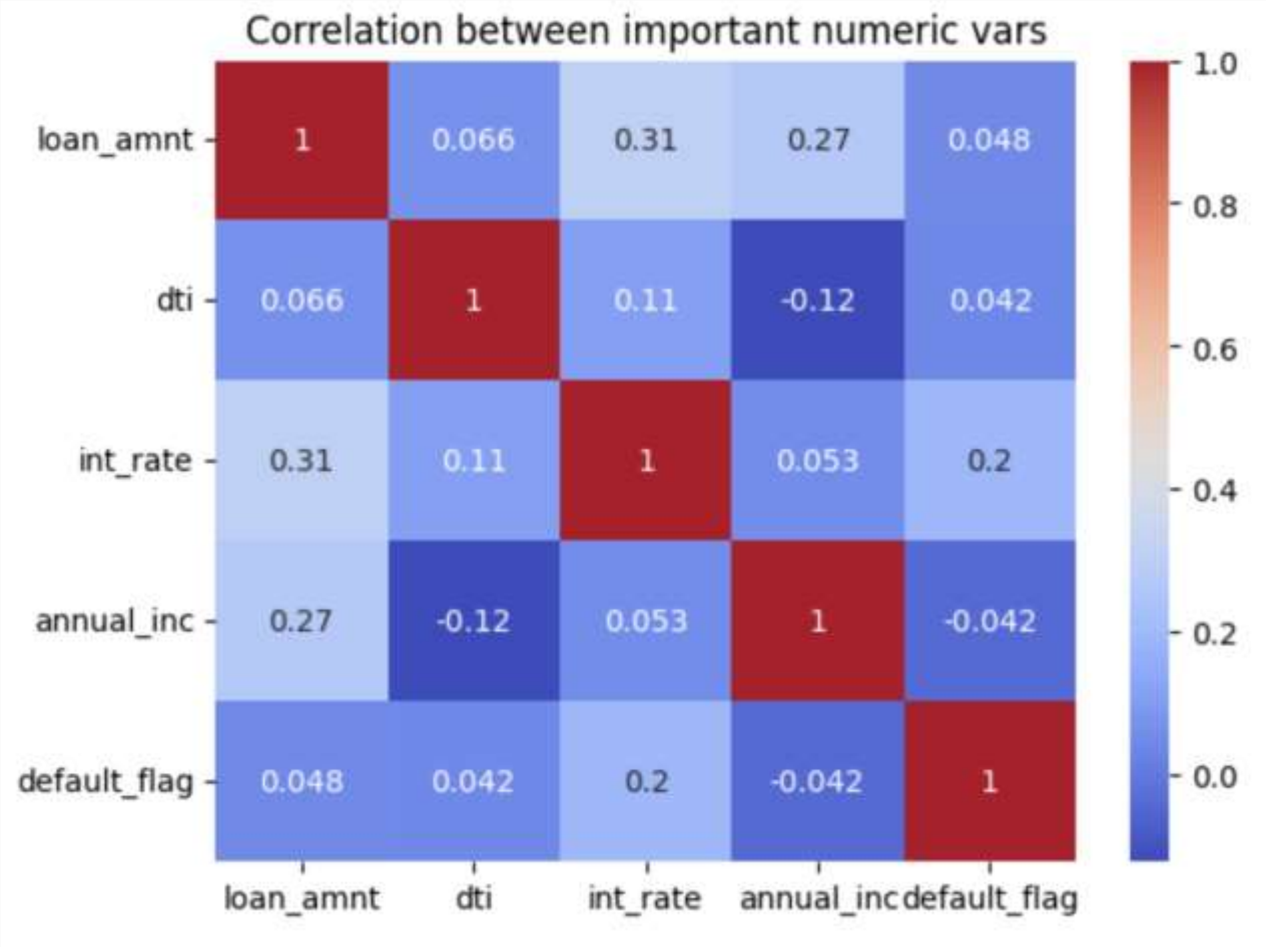


Fig. 2

Another insight from a heatmap (fig2) can be drawn from 4 numerical variables in our loan dataset, namely- Loan Amount, Debt-to-income ratio, interest rate, annual income and default flag.

- Loan amount <-> annual income **=>** Positive correlation. An Expected and healthy correlation
- Interest Rate <->Loan Amount => Positive correlation.
- DTI<-> Default => Positive correlation. Borrowers already in debt are more certain to miss payments
- Annual Income <-> Default => Negative Correlation. Low income applicants are likely to default more.

Conclusion

Company’s loan data set reflects expected financial behavior

Higher income candidates will have higher loan amounts and lower default risk

Higher interest rates for lower grade candidates indicate higher DTI which further means greater probability of default